

Terrain Simulator Final Report

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Abstract—Haptic devices for terrain simulation are becoming increasingly important for applications such as virtual reality and gait training. Typical haptic devices used for these purposes either struggle to completely engage with the human gait pattern or become impractical outside of a research setting. In this paper, we present a 5-bar linkage, 2-DOF, kinesthetic haptic device that straps on to the user's foot and is capable of simulating a variety of objects and virtual environments. We performed a user study with two users, in which we investigated the effects of varying the gain in our impedance controller, as well as the users' ability to distinguish between static environments. The study revealed that the device is capable of simulating distinguishable environments, but it could benefit from additional refinement to the mechanical system in order to enhance the user's experience.

I. INTRODUCTION

Navigating various terrains is an important aspect of human existence. With virtual reality becoming increasingly pervasive in today's world, whether for entertainment or rehabilitation, the need to accurately replicate this natural process in virtual environments is constantly growing. Common methods of controlling human locomotion in virtual reality consist of joysticks, keyboards, and rarely, specialty treadmills. Handheld methods may be effective for specific applications such as gaming, but there is much room for improvement in terms of creating an accessible, immersive, and effective haptic device to complement the locomotion experience in virtual reality. Higher-cost alternatives such as treadmills may be capable of simulating speeds and even certain inclines, but are not able to replicate high-fidelity emulations of dynamic terrains and variable conditions underfoot.

II. BACKGROUND

Haptic devices focused on creating an immersive locomotion experience are continuously being developed. For example, single-DOF devices focused on ankle actuation have proven to be effective for simulating locomotion in virtual reality [1]. However, this type of device is limited in its ability to engage with the entire gait pattern, as the human gait is more complicated than what can be captured by a single-DOF device. Higher DOF devices that include a large support frame have also been developed [2],[3], but these more complicated devices can become quite cumbersome and impractical outside of a research setting. Omnidirectional treadmills have become fairly popular in the context of virtual reality for quite some time [4], but they lack the ability to vary stiffness or topography in a manner that matches real-world terrain. Additionally, devices that mimic ground deformation underneath the foot have been used to

enhance virtual reality experiences [5], but ultimately fail in creating a kinesthetic environment that the user can fully interact with. The terrain simulator we present in this paper aims to be a compact, adaptable, kinesthetic haptic device, capable of rendering a variety of virtual environments that are not limited by their topography, nor the device's range of motion.

III. METHODS

Underfoot sensations to accurately emulate a variety of walking conditions are varied and require a significant amount of variation in force-rendering capabilities. The primary focus of the virtual environments rendered was simulating obstacles, both dynamically and statically for the user to experience and navigate with their feet. Seven successful virtual environments were developed: flat ground, inclined ground, a step/box obstacle, muddy ground, a moving staircase, a vertical wall (with sinusoidal or spring reaction forces), and a series of obstacles that appear in front of the user as they walk forward.

A. Design

Five bar linkages are mechanisms capable of controlling an end effector at any point within its planar workspace. They are often used for pantographs, robotics, and in other haptic devices. The design of a five bar linkage is contingent on the size of the desired workspace, which for the terrain simulator must fully contain normal and abnormal gait conditions for walking humans. To characterize these, we employed a computer vision pose estimation application to plot hip, knee, ankle, and foot coordinates of various walking conditions. From these measurements, we determined that a sufficient workspace to emulate a stride from a high seated position was 14 inches wide and 5 inches tall. Further, the forces conveyed to the user must be chosen which can simulate firm grounds and walls must be determined. Anecdotal trials determined that roughly 10–25 pounds of loading was discernable through the foot. From these two parameters, static/kinematic simulation of a 5 bar linkage/pantograph style 2DOF mechanism informed the decision to use 4.5" lower links, 9" upper links, and a 6" ground link.

B. Electro-Mechanical Design

The motors used were 6374 brushless, sensored outrunner motors commonly used on electric skateboards and capable of producing 5N·m of force, each geared with a 5:1 gear ratio to the crank. This allows the system to operate without overly straining the motor, even in peak torque conditions. To drive the motors, Trampa VESC 6.6 (Vedder Electronic

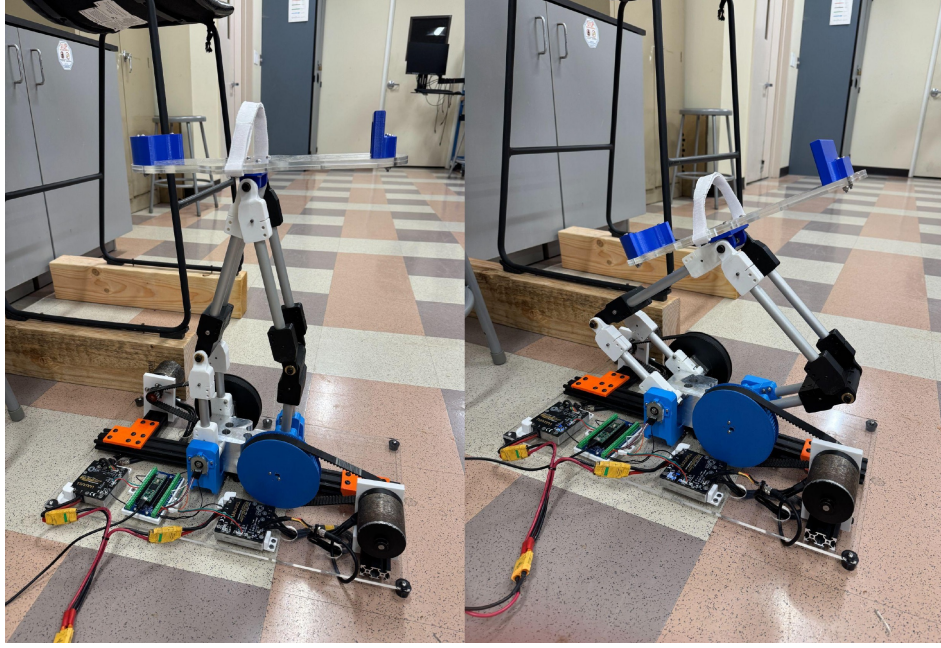


Fig. 1. The device, pictured on the left in the ‘home’ position, with the lower links at 90 degrees to the base link, which it is returned to whenever a new virtual environment is loaded. On the right, the device is pictured in an optimal ‘elbows out’ configuration, where the middle joints are both situated outside of the lower joints, which is important to prevent singularities and inversion of the links which results in breakage. In both images, the end effector (with the foot mount attached) is in roughly the center of the x axis of the devices workspace.

Speed Controllers) is used, due to its ease of setup, integrated PID tuning, and advanced FOC motor control methods, and is connected to a Teensy 4.1 board which renders the virtual environments and conveys current commands to the VESCs. A key benefit of the five-bar linkage is that just two joint angles are enough to define the location of the end effector, thus only two AMT103 quadrature encoders are necessary at the bottom of the linkage to track our system in space.

Mechanically, 20-40 extrusion mounted to an acrylic base plate is used as a base frame for modularity and adjustability, and the links are cut from $\frac{3}{4}$ " thin walled aluminum tubing, capped with 3D printed joints with brass-graphite bushings press fit into place, and pinned together with shoulder screws. A laser cut plate with adjustable end-stops and a velcro strap is used to hold the users foot to the linkage, and is allowed to freely pivot about that joint. Motor mounts, pulleys, encoder mounts, electronics enclosures and 40-80 gussets are 3D printed and comprise the rest of the mechanical system with the exception of the bottom link which was machined from billet aluminum and had bushings press fit in place for the steel drive shaft.

C. Virtual Environment Simulation

The device includes seven virtual environments for the user to interact with, including two dynamic terrains. The static environments consist of flat ground, a box, a wall, a metal grate, and mud. Whereas terrains consisting of repeated boxes that the user can continuously step over and an escalator make up the dynamic terrains. The objects in each of the environments were simulated using impedance control. To simulate horizontal surfaces, such as the ground,

we set our stiffness to 20lb/in, and for vertical surfaces, we implemented a stiffness of 5lb/in. While the difference between these two gains is large, we found that the user could not generate nearly as much force against vertical surfaces as they could against horizontal surfaces, and limiting the gain against vertical surfaces prevented instabilities in the system when the user was simultaneously interacting with both types of objects. Additionally, viscous damping was implemented in the mud environment using a damping gain of $K_d = 0.2\text{lb*s/in}$, and negative damping was implemented in free space to mitigate the inertia felt by the motor using a damping gain of $K_d = -0.025\text{lb*s/in}$. Lastly, the metal grate environment implemented an additional decaying sinusoid to the force generated at the end effector to simulate vibrations upon impact with the wall.

D. User Study

To gather a qualitative metric of the ability of the system to render a wide variety of virtual environments, a small user study was conducted. Two participants were provided with the following list of the names of five static virtual environments: flat ground, ramp, metal grate, stationary box, and wall. The operator of the system chose randomly from this list (without the participants knowledge), and the participant was asked to explore the and identify by name the environment they were experiencing. Per participant, these five trials were repeated for two different impedance gain values ($K_p = 20$, and $K_p = 30$ lbs/in respectively).

The accuracy of each participant, as well as their qualitative observations/comments about their experience of the environments, are provided in the following section.

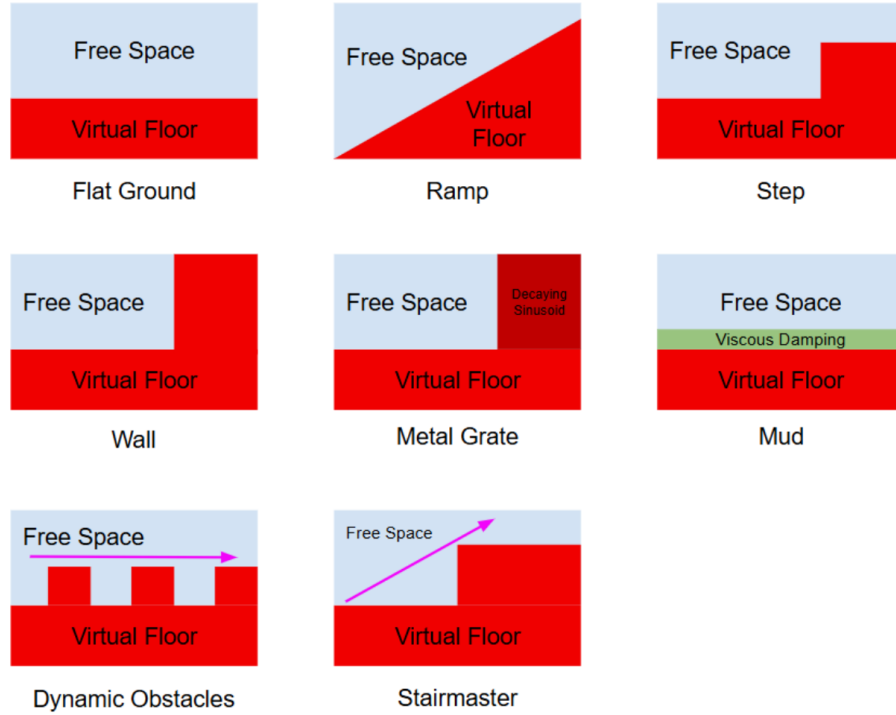


Fig. 2. Representations of the virtual environments rendered by the terrain simulator. Flat ground, ramp, step, wall, and metal grate were rendered as part of the user study, whereas mud, dynamic obstacles, and stairmaster were created to better define the capabilities of the device. The dynamic environments (obstacles and stairmaster) advance the user in the direction of the arrow as they move their foot backward while in contact with the ground plane, providing a rough simulation of one-footed walking. The metal grate environment is physically identical to the wall but includes a decaying sinusoidal response on impact. The mud environment adds viscous damping in both X and Y directions when the foot is near the ground surface.

IV. RESULTS

A. Participant 1

	1	2	3	4	5
Rendered	ramp	step	flat ground	wall	metal grate
Guessed	ramp	flat	metal grate	wall	stationary box

TABLE I

PARTICIPANT 1, TRIAL 1 RESULTS

1) *Trial 1* ($K_p = 20 \text{ lbs/in}$): In trial one's second rendered environment, the participant failed to travel far enough horizontally to identify the step/obstacle in the environment, incorrectly identifying it as flat ground. In the subsequent environment, the participant noted that he felt the sensation of flat ground, but chose metal grate out of a desire to not change his previous answer. The participant also noted that he used the process of elimination to identify the "metal grate" environment as "stationary box," but was not confident in his answer.

	1	2	3	4	5
Rendered	step	wall	ramp	flat ground	metal grate
Guessed	step	metal grate	ramp	flat ground	wall

TABLE II

PARTICIPANT 1, TRIAL 2 RESULTS

2) *Trial 2* ($K_p = 30 \text{ lbs/in}$): Participant one was visibly more comfortable/adept in his use of the device in the second trial. He was able to identify the step, wall, and ramp by finding the boundary between the obstacle and free space, tracing it, and using its shape to inform his estimate of its name. He also clearly became more familiar with the extent of the workspace, no longer reaching the mechanical limits of the device's range of motion. Finally, he noted confusion at the name of the environment "metal grate", which renders an impact vibration proportional to velocity when the user enters a vertical boundary. The participant noted that the impact vibration (and corresponding sound) increased his perception of the rigidity of the boundary, and switched his guesses for metal grate and wall trials, which were previously correct.

B. Participant 2

It is worth noting that participant two had experience with the device from a brief demo prior to the experiment. Additionally, participant two was encouraged to explore the device's range of motion and force rendering capabilities in a short interaction with a basic, flat ground virtual environment before beginning the experiment. A "user calibration" period seemed helpful in allowing the participant to distinguish between the device's mechanical limits and rendered virtual environments, which was a point of confusion for participant one.

1) *Trial 1* ($K_p = 30 \text{ lbs/in}$): Participant two's first trial had an impedance gain of $K_p = 30 \text{ lbs/in}$. He correctly

	1	2	3	4	5
Rendered	ramp	step	flat ground	wall	metal grate
Guessed	ramp	step	flat ground	wall	metal grate

TABLE III
PARTICIPANT 2, TRIAL 1 RESULTS

identified all environments, but noted he struggled to identify the horizontal wall component of the step. He also identified instability/vibration when in contact with an obstacle (which was not intentionally rendered with the environments).

	1	2	3	4	5
Rendered	step	wall	ramp	ground	metal grate
Guessed	wall	flat ground	ramp	flat	metal grate

TABLE IV
PARTICIPANT 2, TRIAL 2 RESULTS

2) *Trial 2* ($K_p = 20 \text{ lbs/in}$): In the trial of the step environment, participant two failed to identify the top of the step, but correctly recognized a horizontal impedance. Interestingly, in the “wall” case, he failed to recognize horizontal impedance. Later, when presented with the flat ground environment, he noted the lack of a vertical obstacle, and realized he previously miscategorized the “wall” environment as flat ground. In all cases, he stated the lower impedance gain felt better and “smoother” than the higher gain of the first trial.

V. DISCUSSION

The distinction between physical limits of the device and rendered virtual walls proved quite important, and hard to delineate without revealing too much about the virtual environment the participant was supposed to be exploring. As a result, certain environments such as wall, flat ground, and metal grate could be easily confused. Conversely, environments that encouraged constant contact with the rendered wall such as the ramp environment proved to have near perfect accuracy among participants. Further, without any sort of visual cue or graphical representation of their environment it was hard for participants to ‘know’ to reach all the way forward and ‘find’ obstacles. When presented with an obstacle or wall early in their first set, participants tended to be more conservative/cautious with their forward extension in later simulations, understandably as their experience is most akin to feeling around blindly with their feet in the dark.

Lower gains were reported to feel overall “better” with less control instability at virtual walls and a smoother mechanical experience. Higher gain does result in stiffer walls, which means a more clearly defined obstacle for the user to experience, however due to the relative instability of our mechanical system, it quickly becomes problematic and detrimental to the user experience.

VI. CONCLUSIONS

Significant haptics research focuses on engagement with users’ hands, and the recreation of manipulating objects and

actions by one’s hands, however, this ignores that human’s feet are always in contact with the ground, regardless of their environment or undertaken action, and if virtual reality is to progress to the next level, technology must be able to simulate the basic sensations of our feet as users stand, walk, and run on various surfaces. By utilizing a 5 bar linkage, we demonstrated effective simulation of a variety of virtual environments and conditions underfoot with a system designed to output up to 25 pounds of force to the user. Unfortunately, rapid manufacturing methods such as 3D printing result in durability and strength limits, and restrict the system from reaching its full torque and force potential before part failure have limited the system to a maximum of 15 pounds of simulated force. However, just 15 pounds resulted in very stiff walls and floors, well detectable and identifiable by users, and the strong gearing and motors did not compromise free-space fidelity. This indicates that future work— most significantly, building a second device for the user’s other foot— is quite necessary to explore the feasibility of rendering more complex environments and accurate sensation. Ideally, a device such as this should be manufactured with enough strength and motor torque to support the users entire weight, removing the need for a chair or harness to support most of their weight and resulting in a truly untethered haptic experience.

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